DEV TYAGI

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FPGA/RTL Design Engineer – RISC-V Microarchitecture – AI Accelerator Design – AXI4 Verification

Summary

FPGA/RTL design engineer specializing in RISC-V microarchitecture, AXI4 memory subsystems, and AI accelerator datapaths on Xilinx Zynq/UltraScale+. SystemVerilog/UVM verification background with proven results: 250 MHz post-synthesis timing closure, >95% functional coverage, 1 GB/s sustained DMA throughput.

Technical Skills

HDL / Verification: SystemVerilog, Verilog, VHDL, SVA, UVM, Constrained-Random Verification, Functional Coverage, Scoreboards
RTL / Microarchitecture: Out-of-Order Execution, Tomasulo's Algorithm, Superscalar Pipelines, Hazard Resolution, Clock Gating, CDC
ASIC / FPGA Flow: RTL Synthesis, Place & Route, STA, Timing Closure, Lint, DRC, Equivalence Checking
Protocols: AXI4, AXI4-Lite, AXI4-Stream, AHB, SPI, I2C, UART
Platforms: Xilinx Zynq-7000, UltraScale+, ARM Cortex-A9 (PS/PL Co-design), PYNQ-Z2
Tools / Languages: Vivado, ModelSim, Questa Sim, VCS, Xcelium, Git, Tcl, C/C++, Python, Embedded C

Experience

FPGA RTL Design Intern — *Vicharak (Remote)* Mar 2026 – Present

- Ported PicoRV32 RISC-V core to Shrike Lite FPGA, fitting execution within tight LUT/BRAM budgets the original configuration exceeded.
- Replaced AXI4-Lite memory interface with a custom SPI protocol bridge for off-chip flash-backed fetch, cutting fabric logic utilization by over 30%; authored SPI controller and memory-mapped bridge RTL from scratch in SystemVerilog.

RTL Design Intern — *IIT Palakkad* May 2026 – Jul 2026

- Designed RTL for a neural network inference accelerator on Xilinx Zynq-7000, mapping convolutional and fully-connected layer datapaths into PL with Cortex-A9 PS handling control flow.
- Built AXI4-MM/AXI4-Stream interfaces for low-latency weight/activation DMA; implemented INT8/INT16 fixed-point datapaths on DSP/BRAM resources, holding inference accuracy within target bounds.

Team Lead, FPGA Signal Processing — *Army Design Bureau Collaboration, VIT Chennai* Jan 2026 – Present

- Built real-time DSP pipelines (FFT, FIR filtering, feature extraction) in Zynq PL for RF signal classification, with ARM Cortex-A9 PS coupled over AXI4/AXI-Stream.
- Cut end-to-end system latency by 15% through RTL pipelining, CDC fixes, and datapath restructuring; led RTL reviews and integration testing across a 40+ member team.

Projects

Out-of-Order Superscalar RISC-V Processor — *SystemVerilog, Vivado, ModelSim* devtyagi3909/riscv-ooo-core

- Full superscalar RV32I OOO processor using Tomasulo's Algorithm – reservation stations, ROB, CDB – with a 7-stage pipeline supporting speculative execution and precise exception handling.
- >95% functional coverage via SVA property checks and constrained-random stimulus; 250 MHz timing closure on Xilinx UltraScale+ post-synthesis; passed RISC-V ISA compliance tests. 1st Place + Special Jury Award, SanDisk Hardware Hackathon (100+ teams).

PMDC Motor Drive FPGA Simulation — *SystemVerilog, Zynq-7000, lwIP, Python* devtyagi3909/pmdc-fpga-motor-control

- Deployed closed-loop PMDC motor drive simulation RTL on Zybo Z7-20 (Zynq-7000 PL fabric), including an internally-generated PWM chopper block with multi-frequency preset switching.
- Streamed real-time motor telemetry over TCP via AXI DMA and the lwIP stack to a Python live dashboard for closed-loop monitoring and control validation.

High-Throughput AXI4 DMA Engine — *SystemVerilog, Vivado, PYNQ-Z2*

- AXI4-MM/AXI4-Stream DMA engine with configurable burst lengths and an interrupt-driven descriptor architecture; achieved 1 GB/s sustained throughput on PYNQ-Z2 under real AXI traffic, verified with SVA and constrained-random testing. 2nd Place, FPGA Makeathon.

ConstraintForge – Open-Source FPGA Timing Constraint Library — *Tcl, Python, GitHub Actions* devtyagi3909/constraintforge

- Annotated XDC/SDC/LPF constraint templates covering 14+ interfaces (AXI4, SPI, DDR3, RGMII, HDMI, MIPI CSI-2) across Xilinx, Intel, Lattice, and Gowin; every file CI-validated via GitHub Actions.

Achievements & Leadership

- 1st Place + Special Jury Award, SanDisk Hardware Hackathon (RTL Track, 100+ teams) – out-of-order RISC-V processor design.
- 2nd Place, FPGA Makeathon – AXI4 DMA engine sustaining 1 GB/s throughput on PYNQ-Z2.
- Head, Robotics Club, VIT Chennai – led 40+ engineers across hardware development, workshops, and intercollegiate competitions.

Education

B.Tech, Electronics and Computer Engineering — Vellore Institute of Technology, Chennai Aug 2023 – May 2027